

Q1) Long Weekend



Figure 1: Fall in Laurentian Mountains

Exercises:

- 35 → Adult food charges for the day will be: \$55
'Y' 'N' 'Y'
- 5 → Toddler food charges for the day will be: \$5
'y' 'y' 'y'
- 12 → Child food charges for the day will be: \$20
'n' 'N' 'Y'

Code:

```
charges = 0

if age > 12:
    if b == "Y" or b == "y":
        charges += 20
    if l == "Y" or l == "y":
        charges += 30
    if d == "Y" or d == "y":
        charges += 35
elif age > 5 and age < 13:
    if b == "Y" or b == "y":
        charges += 10
    if l == "Y" or l == "y":
        charges += 20
    if d == "Y" or d == "y":
        charges += 20
elif age <= 5:
    if d == "Y" or d == "y":
        charges += 5

if age > 12:
    print("Adult food charges for the day will be: $" + str(charges))
elif age > 5 and age < 13:
    print("Child food charges for the day will be: $" + str(charges))
elif age <= 5:
    print("Toddler food charges for the day will be: $" + str(charges))
```

Q2) Chocolate Distribution



Figure 2: Chocolates

Exercises:

- 5 → Professor Mars will need to buy 20 chocolates to distribute in class
3 2
6 4
9 6
12 8
1 10
- 7 → Professor Mars will need to buy 22 chocolates to distribute in class
2 15
3 6
5 12
6 3
8 9
9 1
1 1
- 3 → Professor Mars will need to buy 15 chocolates to distribute in class
5 5
6 6
7 7

Code:

```
count_choc = 0
for i in range(N):
    n = int(input())
    m = int(input())
    performance = n/m * 100

    if (performance < 40):
        count_choc += 0
    elif performance >= 40 and performance < 50:
        count_choc += 1
    elif performance >= 50 and performance < 65:
        count_choc += 2
    elif performance >= 65 and performance < 80:
        count_choc += 3
    elif performance >= 80 and performance < 100:
        count_choc += 4
    else:
        count_choc += 5
    #print(i, count_choc)

print("Professor Mars will need to buy " +
      str(count_choc) + " chocolates to distribute in class")
```

Q3) Matherns

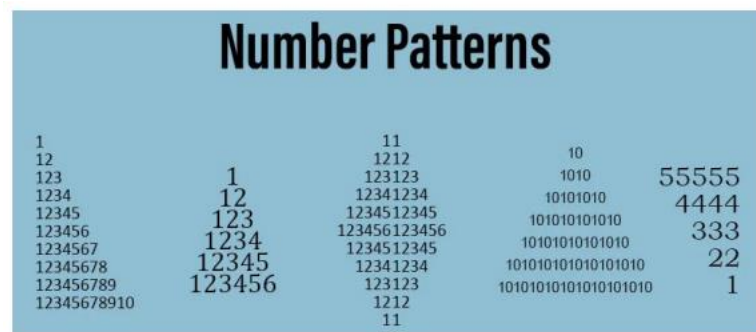


Figure 3: Patterns are FUN!

Exercises:

- 4
 - 4 16 64 256
 - 3 9 27
 - 2 4
 - 1
- 7
 - 7 49 343 2401 16807 117649 823543
 - 6 36 216 1296 7776 46656
 - 5 25 125 625 3125
 - 4 16 64 256
 - 3 9 27
 - 2 4
 - 1
- 2
 - 2 4
 - 1

Code:

```
for i in range(n, 0, -1):
    for j in range(1, i+1):
        print(i**j, end=' ')
    print()
```

Q4) Golden Wedding



The Weasleys



Hogwarts Lake

Figure 4: Golden Wedding Anniversary of Mr. & Mrs. Weasley

Exercises:

- 1 2 3
 - Rooms required: 6
 - All the rooms will cost: £3213.18
- 7 9 5
 - Rooms required: 21
 - All the rooms will cost: £9249.08
- 2 6 8
 - Rooms required: 16
 - All the rooms will cost: £8728.64

Code:

```
def GetRoomRate(room_code):
    if(room_code == "1PR"):
        room_rate = 115.11
    elif (room_code == "SFR"):
        room_rate = 235.23
    elif (room_code == "LFR"):
        room_rate = 340.34
    return room_rate

def GetRoomCost(num_rooms, room_code):
    room_rate = GetRoomRate(room_code)
    room_cost = room_rate * num_rooms
    return room_cost

s = int(input())
sf = int(input())
lf = int(input())

cost_individual = GetRoomCost(s, "1PR") * 2
cost_small_family = GetRoomCost(sf, "SFR") * 2
cost_large_family = GetRoomCost(lf, "LFR") * 2

total_rooms = s + sf + lf
total_room_cost = round(
    cost_individual + cost_small_family + cost_large_family, 2)
print("Rooms required: " + str(total_rooms))
print("All the rooms will cost: £" + str(total_room_cost))
```

Q5) Debugging

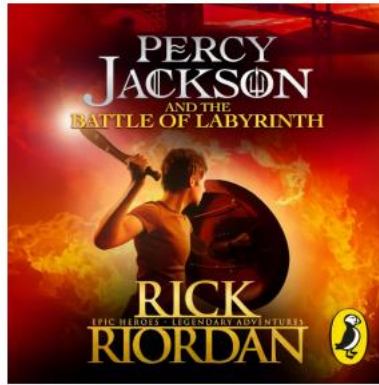


Figure 5: Percy Jackson & the Battle of Labyrinth

Errors:

- Line 2: Incorrect function name. The question says “solving_labyrinth” and the code says “solve_labyrinth”
- Line 2: Incorrect argument to the function – should be ‘n’ instead of ‘N’, because the entire function body uses ‘n’ in its execution
- Between line 2 and line 3: Variables sum_even and sum_odd must be defined before using them in the function
- Line 3: Incorrect syntax for loop variable – cannot start with a number
- Line 4: Use num1 as the rest of the program uses that for calculation
- Line 8: Calculation for odd numbers is missing
- Line 9: Incorrect calculation for average
- Between line 10 and line 11: odd sum needs to be printed after even sum
- Line 11: The average variable is incorrect – should be “average” instead of “avg”
- Line 13: No function call to run the code

Correct code:

```
def solve_labyrinth(n):
    sum_even = 0
    sum_odd = 0
    for i in range(n):
        num1 = int(input())

        if (num1 % 2 == 0):
            sum_even += num1
        else:
            sum_odd += num1

    average = (sum_even + sum_odd)/2
    print(sum_even)
    print(sum_odd)
    print(average)

solve_labyrinth(n)
```